

User Manual

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0.2	2025-12-23	Updated to match GUI v2 layout and user actions	AF, AG, VG, MM	DN

1 INTRODUCTION

This document is the **User Manual** for the **Automatic Music Transcription (AMT) System** developed in the course **TSKS23: Project Course in Signal Processing, Communications and Networking (CDIO)** at Linköping University.

The AMT system allows a user to:

- record piano audio in real time via microphone, or
- load an existing audio file,

and automatically produce a transcription consisting of note names and timing information (onset, offset, duration). Results are visualised (waveform + spectrogram), can be exported, and are stored locally for later inspection.

This manual is written for end users and should be sufficient to install, start, and use the system without assistance.

2 SYSTEM REQUIREMENTS

2.1 Hardware

- A computer with microphone input (for live recording)
- Piano/keyboard (system targets 61-key range C2–C7)
- Recommended: quiet recording environment for best accuracy

2.2 Software

- Operating system: Windows or Linux (macOS may require extra audio setup)
- Python: 3.10 or later
- Required packages are installed from `requirements.txt`

3 INSTALLATION AND SETUP

3.1 Install Dependencies

1. Download or clone the project repository.
2. Open a terminal in the project folder.
3. Install required Python packages:

```
pip install -r requirements.txt
```

3.2 Audio Recording Notes (PyAudio)

If live recording does not work, the most common cause is that PyAudio/PortAudio is missing or misconfigured. Ensure your microphone is enabled in your operating system settings and that the correct audio input device is selected.

3.3 Start the Application

Run:

```
python main.py
```

This opens the graphical user interface (GUI).

4 SYSTEM OVERVIEW

The system consists of five main modules:

1. Input Handling System (file loading + live capture)
2. Graphical User Interface (GUI)
3. Backend Processing Engine (onset/pitch detection + transcription)
4. Database Management System (local storage)
5. Output and Export Module (display + exports)

The GUI provides controls and visualisations. The backend performs signal processing and returns transcription results, which are then displayed and optionally exported/stored.

5 GRAPHICAL USER INTERFACE

5.1 GUI Layout

The GUI is organised into two main areas:

- **Left panel:** file selection, live recording controls, transcription settings, and results/export buttons.
- **Right panel:** visualisations (tabs for waveform and spectrogram).

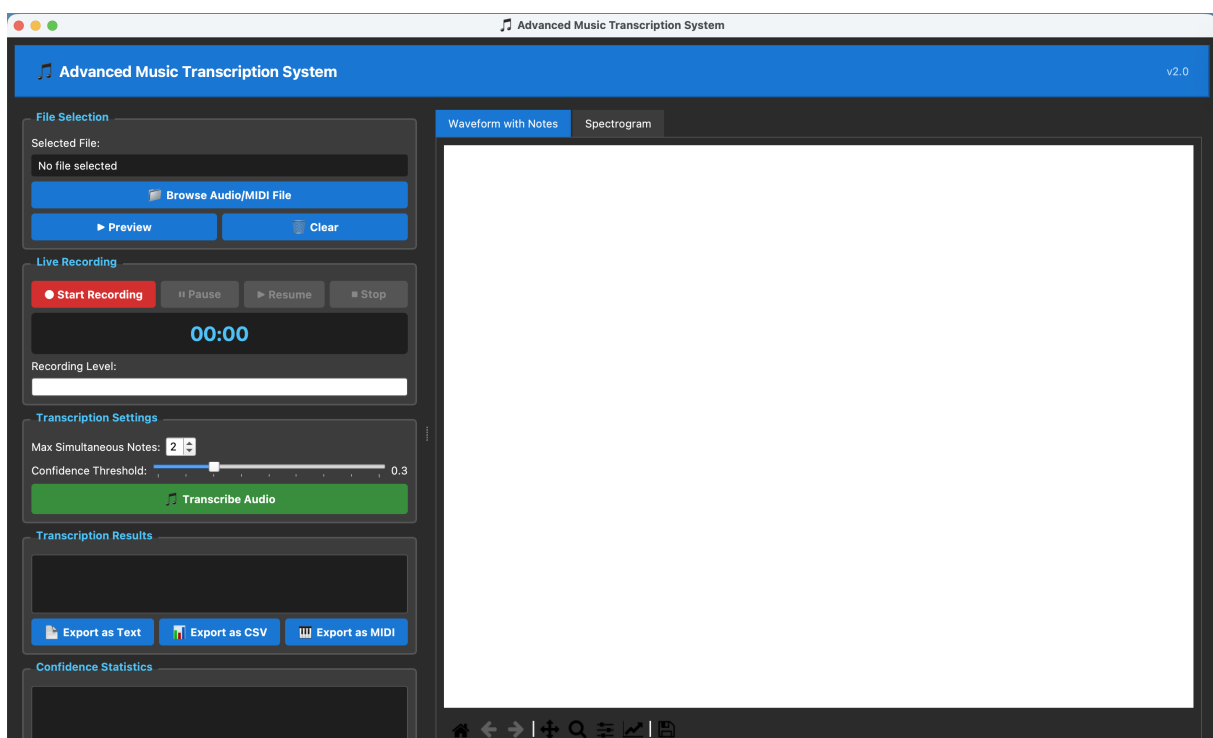


Figure 1: AMT System GUI (example).

5.2 File Selection Controls

- **Browse Audio/MIDI File:** opens a file picker to select an input file.
- **Preview:** plays back the selected audio (if enabled in the build).
- **Clear:** clears the current selection and resets the file state.

5.3 Live Recording Controls

- **Start Recording:** begins capturing microphone input.
- **Pause:** temporarily pauses capture (recording session remains active).
- **Resume:** continues capture after a pause.
- **Stop:** ends recording and finalises the captured audio for transcription.

The recording timer shows elapsed time. The “Recording Level” field indicates whether input energy is detected.

5.4 Transcription Settings

- **Max Simultaneous Notes:** limits how many notes the system reports per event (used to control chord density).
- **Confidence Threshold:** filters out low-confidence detections. Increasing it produces fewer notes; decreasing it produces more notes (and potentially more false positives).

5.5 Transcription Results and Exports

The **Transcription Results** area displays the output note list after transcription.

Export buttons (only use what your implementation supports):

- **Export as Text:** saves results to a `.txt` report.
- **Export as CSV:** saves results to `.csv` format for spreadsheet analysis.
- **Export as MIDI:** saves a `.mid` file that can be opened in music software.

6 TYPICAL WORKFLOWS

6.1 Workflow A: File-Based Transcription (Offline)

1. Click **Browse Audio/MIDI File** and select a file.
2. (Optional) Click **Preview** to listen.
3. Set **Max Simultaneous Notes** and **Confidence Threshold**.
4. Click **Transcribe Audio**.
5. Inspect the note list and visualisations.
6. Export results if needed.

6.2 Workflow B: Live Recording Transcription

1. Ensure your microphone is connected and enabled by the OS.
2. Click **Start Recording**.
3. Play notes/chords on the piano/keyboard near the microphone.
4. Use **Pause/Resume** if needed.
5. Click **Stop** to end recording.
6. Click **Transcribe Audio** (or transcription runs automatically depending on build).
7. Inspect and export results.

7 OUTPUT AND STORAGE

7.1 Displayed Output

The transcription output typically includes:

- Note name (e.g. C4, G#5)
- Onset time (s)
- Offset time (s)
- Duration (s)

Results are shown as text and visually on waveform/spectrogram plots.

7.2 Database Storage

Transcriptions and metadata are stored automatically in a local SQLite database in the project directory (or configured output folder). No manual database actions are required by the user.

8 TROUBLESHOOTING

Issue	What to do
No audio detected	Check microphone permissions and OS input device selection. Test microphone in system settings.
Recording fails / PyAudio error	Reinstall dependencies. Confirm PortAudio/PyAudio is installed correctly for your OS.
No notes detected	Lower the confidence threshold, increase input volume, and record in a quieter room.
Too many wrong notes	Increase confidence threshold and reduce background noise. Use single-note tests first.
Export fails	Ensure you have write permission to the chosen folder and that the file is not open in another program.

9 LIMITATIONS AND KNOWN ISSUES

- The system is optimised for piano/keyboard input in the target range C2–C7.
- Accuracy may decrease for dense chords, reverberant rooms, or noisy recordings.
- Results depend strongly on microphone placement and recording quality.

10 BEST PRACTICES

1. Record in a quiet environment and avoid background music/noise.
2. Place the microphone close to the instrument (but avoid clipping).
3. Start testing with single notes, then simple chords.
4. If you see many false notes, increase the confidence threshold.